NETWORKING

The OSI Model Visual Guide

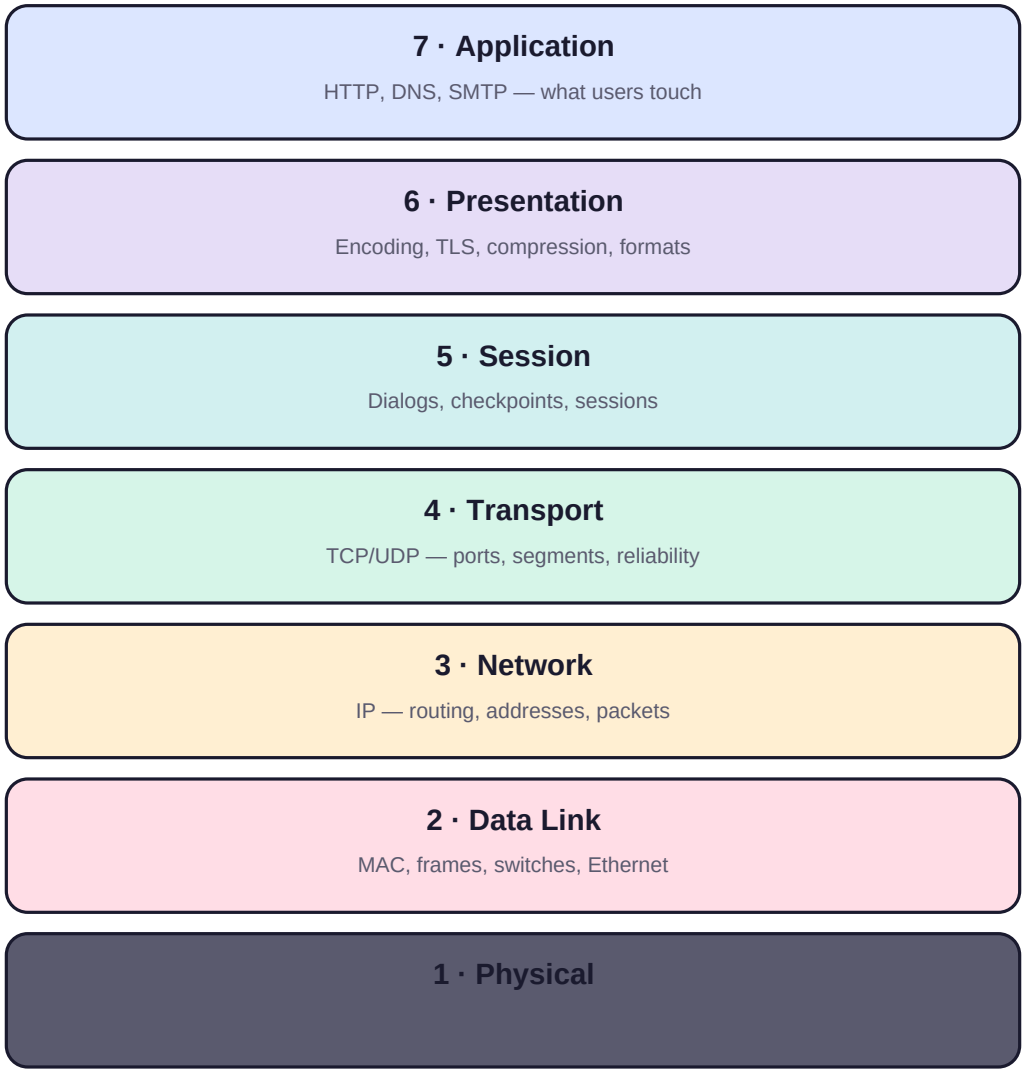
Seven layers of networking —
what each layer does and where the work happens.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The 7 Layers, Top to Bottom

Upper layers = apps & data meaning. Lower layers = moving bits.

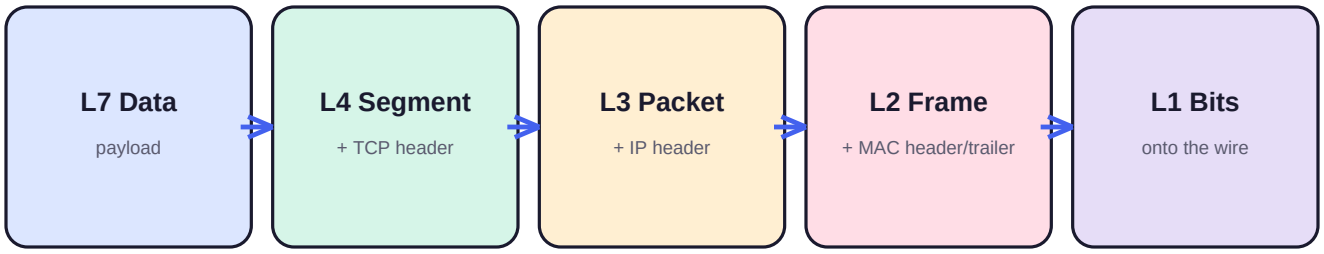


Mnemonic: 'All People Seem To Need Data Processing' (7→1). Most real work happens at 2, 3, 4, and 7.

STEP 2/6

Encapsulation: Headers All the Way Down

Each layer wraps the payload from above and adds its own header.



- Sending: Application data descends; every layer ADDS control info (headers, sometimes trailers).
- Receiving: the peer's matching layer reads and STRIPS its own header, then passes up.
- The unit changes name per layer: message → segment → packet → frame → bits.

STEP 3/6

What Each Header Carries

Control information is how layers coordinate with their peer.

L2: MAC header/trailer

src/dst MAC, frame check sequence (CRC)

L3: IP header

src/dst IP, TTL, protocol — routes the packet

L4: TCP header

src/dst ports, seq/ack numbers, flags, window

L5-7: app headers

HTTP headers, TLS records, MIME types — app semantics

Overhead is real: a tiny payload can carry 40+ bytes of TCP/IP headers plus Ethernet framing. Header compression (ROHC, HPACK) exists for a reason.

STEP 4/6

Decapsulation on Arrival

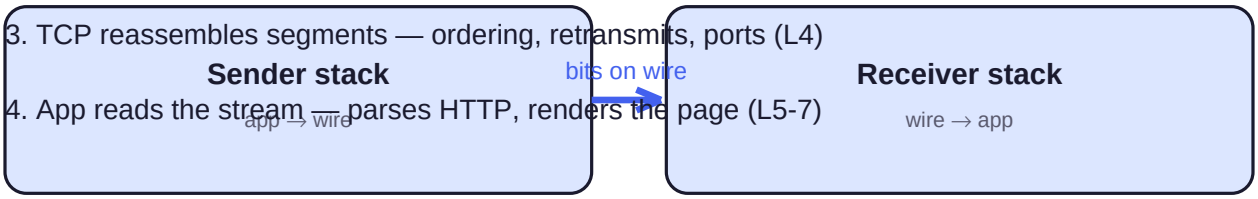
The reverse trip: strip, check, pass up.

1. NIC receives frame — CRC checked, MAC trailer stripped (L2)

2. IP stack reads header — dst IP matches? TTL decremented (L3)

3. TCP reassembles segments — ordering, retransmits, ports (L4)

4. App reads the stream — parses HTTP, renders the page (L5-7)



A device only processes the layers it needs: a switch stops at L2, a router at L3, a firewall often at L4 — end hosts go all the way up.

STEP 5/6

Which Devices Live Where

Networking gear is named by the highest layer it understands.

Hub / Repeater (L1)

Repeats bits. One collision domain — obsolete.

Switch (L2)

Forwards frames by MAC table. One broadcast domain per VLAN.

Router (L3)

Forwards packets by IP routing table. Connects networks.

L4+ devices

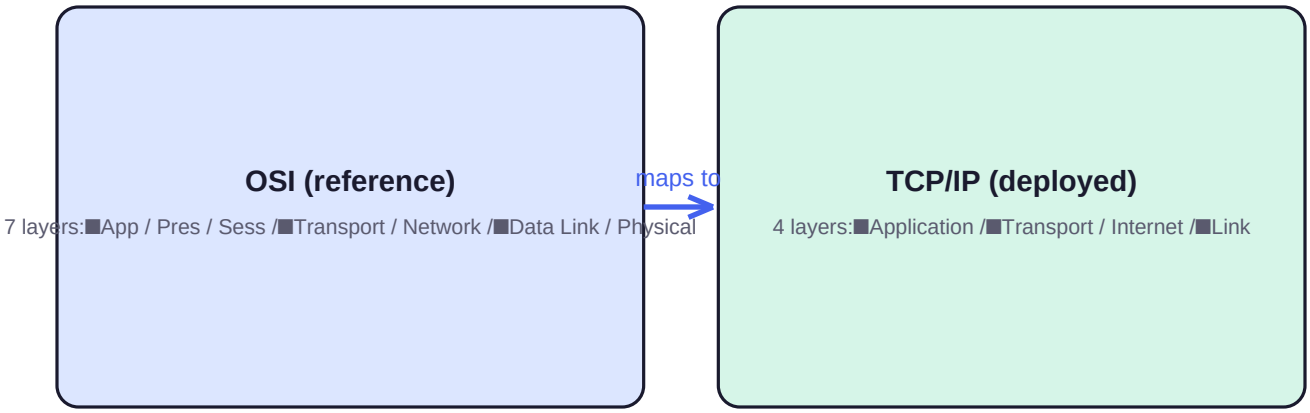
Firewalls, load balancers, proxies — ports, sessions, content.

Rule of thumb: a device that must READ a layer's header to do its job operates at that layer.

STEP 6/6

OSI vs the Real TCP/IP Stack

The internet runs TCP/IP — a leaner 4-layer cousin.



- OSI is a teaching/reference model; TCP/IP is what actually ships.
- TCP/IP folds 5-7 into 'Application' and 1-2 into 'Link' — same ideas, fewer boxes.
- Engineers still say 'L3 switch' or 'L7 load balancer' — OSI numbering is the shared vocabulary.

STEP ✓I✓

Key Takeaways

OSI Model — Remember

- 7 layers: App, Presentation, Session, Transport, Network, Data Link, Physical.
- Encapsulation: every layer adds its header on send, strips it on receive.
- Units rename per layer: message → segment → packet → frame → bits.
- Devices are named by the highest layer they read: switch (L2), router (L3).
- The internet runs TCP/IP's 4 layers, but OSI numbering is the shared vocabulary.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev